IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Devendranath
Chowdary Maganti
9/1/2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

Historical SpaceX Falcon 9 launch data was collected and analyzed using Python, SQL, and visualization techniques. Exploratory Data Analysis (EDA) was performed to identify trends and relationships among payload mass, launch site, orbit type, and landing outcomes. Interactive visualizations were created using Matplotlib, Plotly, Folium maps, and a Plotly Dash dashboard to enable dynamic analysis of launch performance.

- Summary of all results

The analysis shows that landing success rates improve over time, indicating strong learning and technological advancement by SpaceX. Launch success is influenced by payload mass, orbit type, and launch site, with certain sites and lower-complexity orbits achieving higher success rates. Interactive dashboards and maps further highlight spatial and operational patterns, demonstrating that historical launch data can effectively support prediction and decision-making for future missions.

Introduction

- Project background and context

SpaceX has significantly reduced the cost of space launches by developing reusable Falcon 9 rocket boosters. Predicting whether the first stage of a Falcon 9 rocket will successfully land is critical for estimating launch costs and improving mission planning. This project analyzes historical SpaceX launch data to understand factors that influence landing success.

- Problems you want to find answers

How do payload mass, launch site, and orbit type affect first-stage landing success?

Does launch success improve over time as SpaceX gains operational experience?

Which launch sites and mission profiles have the highest success rates?

Can historical data be used to predict the probability of a successful landing?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Web Scraping via BeautifulSoup & Requests from Wikipedia Launch Tables
- Perform data wrangling
  - Cleaned data, encoded categories, created target variable Class (1 = Success, 0 = failure)
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Built, tuned and evaluated 4 models, Decision Tree being the best

Data Collection

- Data Source: Wikipedia – "List of Falcon 9 and Falcon Heavy launches" (en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)
- Key Steps:

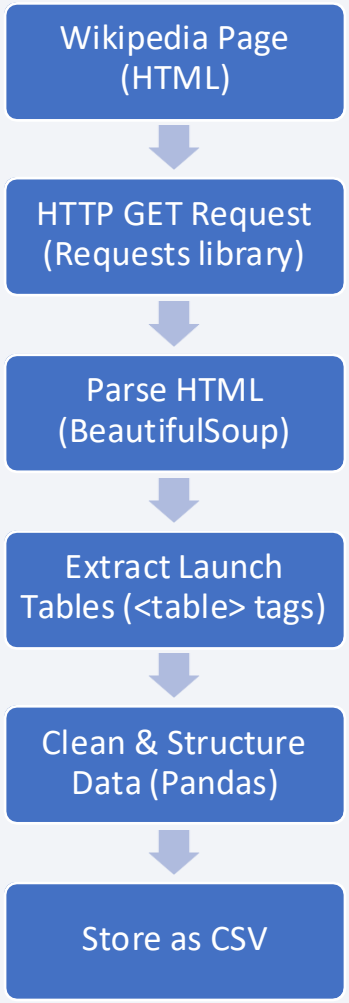
HTTP GET request to fetch live HTML page

Locate tables using HTML tags and class names

Extract rows and columns from each launch record

Clean data – remove footnotes, handle missing values, standardize formatting
Export structured dataset for further analysis

- Tools Used: Python requests – fetch page content; BeautifulSoup – parse HTML; pandas – data structuring and cleaning
- Output: 90+ launch records with fields: Flight Number, Date, Booster Version, Payload Mass, Orbit, Launch Site, Outcome, Landing Status.



Data Collection – SpaceX API

- **Data Source:**
  SpaceX REST API – Launch history and rocket details
  api.spacexdata.com/v4
- Key API Endpoints Used:

  /launches/past – historical launch data

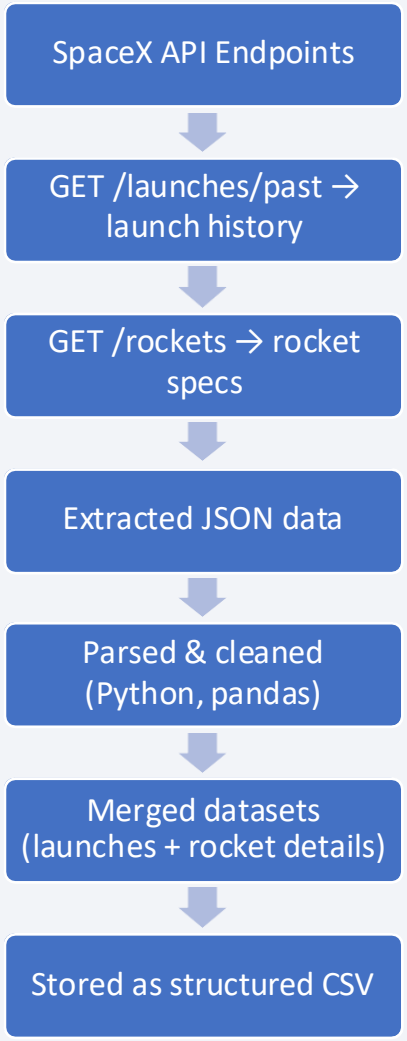
  /rockets – Falcon 9 specifications

  /launchpads – launch site details
- Tools Used:

  Python requests – API calls

  pandas – JSON to DataFrame conversion

  json – parsing API responses
- https://github.com/devendranath997/SpaceX-Landing-Prediction



Data Collection - Scraping

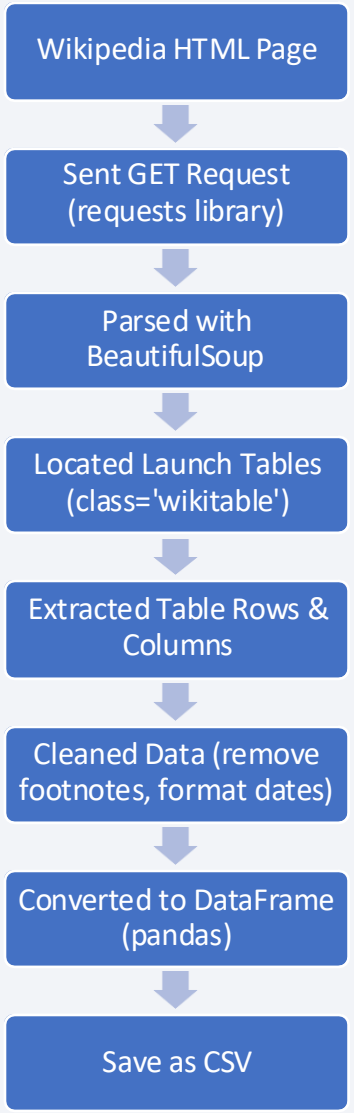
- Tools Used:

requests – fetch HTML content

BeautifulSoup – parse and navigate HTML

pandas – structure and clean data

- https://github.com/devendranath997/SpaceX-Landing-Prediction



Data Wrangling

- Key Steps:

Missing Value Analysis – 28.9% missing in LandingPad

Categorical Encoding – Convert Orbit, LaunchSite, Outcome to numeric

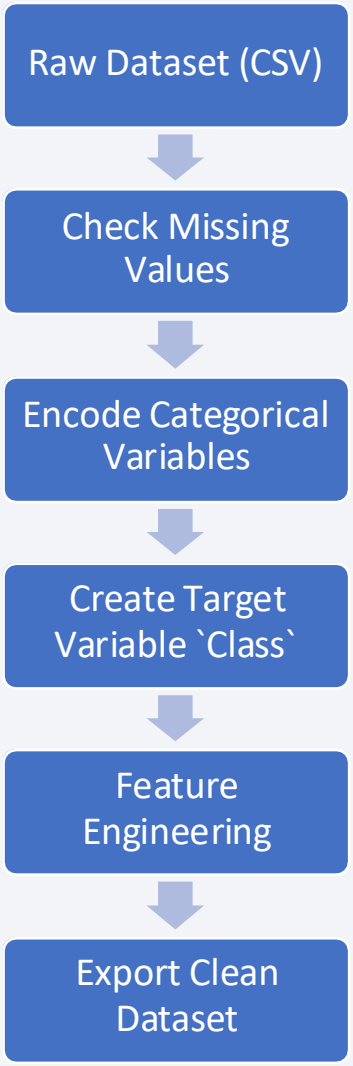
Target Creation – Binary Class column:

1 = Successful landing (True ASDS, True RTLS, True Ocean)

0 = Failed landing (False ASDS, None None, etc.)

Success Rate Calculation – Overall 33.3% landing success

- https://github.com/devendranath997/SpaceX-Landing-Prediction



EDA with Data Visualization

- Scatter plots were used to analyze relationships between flight number, payload mass, launch site, and orbit type with landing outcomes. These plots help identify performance trends over time, capacity limits, and differences across mission conditions.
- Bar charts were used to compare average landing success rates across orbit types, enabling clear comparison of categorical variables and highlighting which mission profiles are more challenging.
- Line charts were used to visualize the temporal trend of landing success by year, demonstrating improvements in mission reliability due to technological learning and operational experience.
- Overall, these visualizations support pattern detection, comparison across categories, and trend analysis, which are core objectives of Exploratory Data Analysis (EDA).
- https://github.com/devendranath997/SpaceX-Landing-Prediction

EDA with SQL

- SELECT and DISTINCT queries were used to explore the dataset structure and identify unique values such as launch sites, orbit types, and mission outcomes.
- Filtering queries (WHERE clauses) were used to analyze specific mission conditions, including successful landings, particular orbit types, and launches from specific sites.
- Aggregate functions (COUNT, AVG, SUM) were applied to compute mission frequencies and calculate average landing success rates across different launch sites and orbit categories.
- GROUP BY queries were used to summarize performance metrics by launch site and orbit type, enabling comparison of mission success across categorical variables.
- ORDER BY and LIMIT clauses were used to rank results and identify top-performing launch sites and the most frequent mission profiles.
- Overall, SQL-based EDA enabled efficient data filtering, aggregation, and pattern discovery directly within the database before visualization.
- https://github.com/devendranath997/SpaceX-Landing-Prediction

Build an Interactive Map with Folium

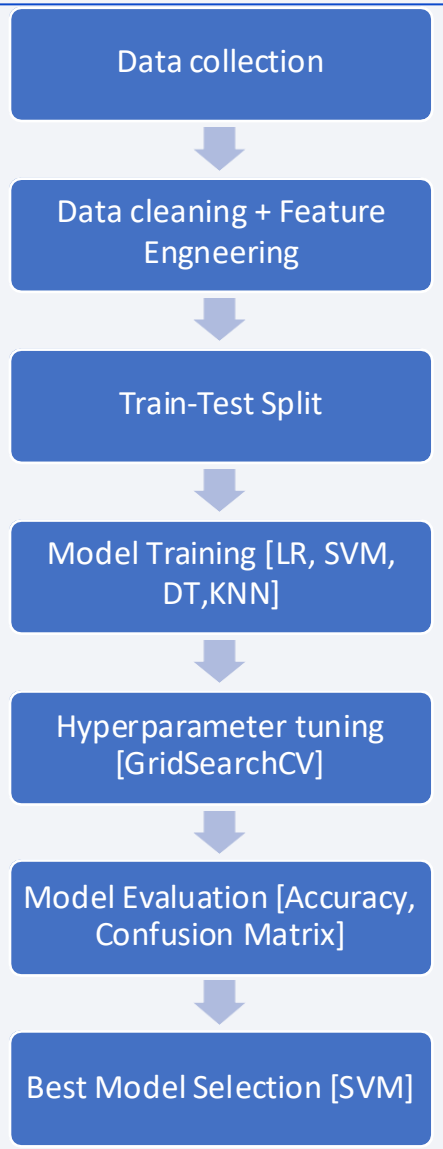
- Markers were added to the Folium map to represent SpaceX launch sites. Each marker includes a popup displaying the launch site name and location, allowing users to interactively explore site-level information.
- Circle markers were used to visualize launch site proximity and to distinguish launch locations from surrounding geographical features.
- Polyline objects were added to show distances between launch sites and nearby landmarks such as coastlines, cities, railways, and highways. This helps assess safety and logistical considerations for launches.
- Color-coded markers were used to differentiate successful and unsuccessful landing outcomes, making spatial performance patterns easy to interpret.
- These map objects collectively enable interactive spatial analysis, helping identify why launch sites are located near coastlines, far from dense populations, and close to transportation infrastructure.
- https://github.com/devendranath997/SpaceX-Landing-Prediction

Build a Dashboard with Plotly Dash

- A dropdown menu was added to allow users to select a specific launch site or view data for all sites. This enables interactive filtering of launch performance by location.
- A pie chart was included to visualize launch success:
- When All Sites is selected, the chart shows total successful launches by site.
- When a specific site is selected, it displays success vs. failure counts for that site.
- This provides an intuitive comparison of launch reliability.
- A payload range slider was added to allow users to filter launches based on payload mass, enabling focused analysis of different payload ranges.
- A scatter plot was used to show the relationship between payload mass and launch outcome, color-coded by booster version category. This helps identify correlations between payload size, booster type, and launch success.
- These interactive components collectively allow dynamic, user-driven exploration of SpaceX launch performance across sites and payload conditions.
- https://github.com/devendranath997/SpaceX-Landing-Prediction

Predictive Analysis (Classification)

- The dataset was preprocessed and feature-engineered, including one-hot encoding of categorical variables and standardization of numerical features.
- The data was split into training and testing sets to ensure unbiased evaluation of model performance.
- Multiple classification models were trained, including Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
- Hyperparameter tuning using GridSearchCV was applied to each model to improve performance and reduce overfitting.
- Models were evaluated using test accuracy and confusion matrices, and Support Vector Machine (SVM) achieved the best (or joint-best) classification performance, making it the final selected model.
- https://github.com/devendranath997/SpaceX-Landing-Prediction



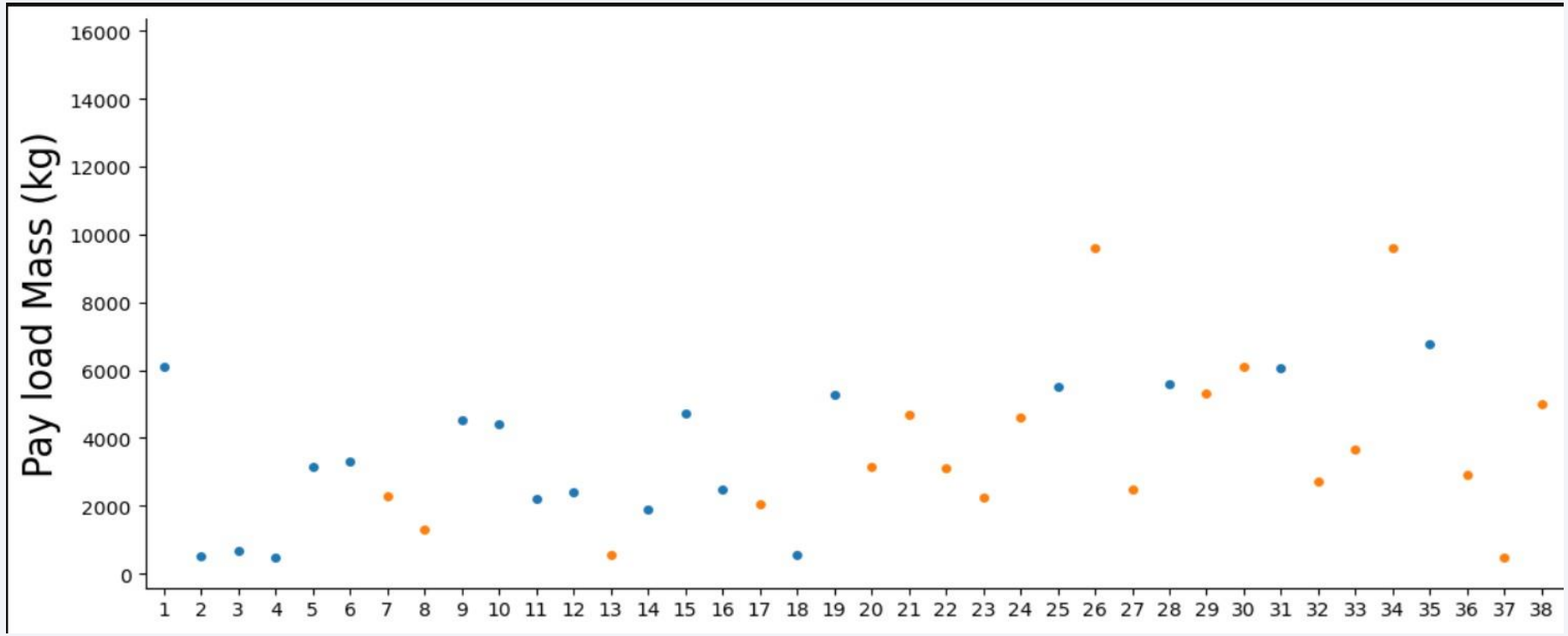
Results

- Exploratory data analysis results
  Analysis showed a clear improvement in landing success over time, with success rates varying by orbit type, launch site, and payload mass. Earlier missions exhibited lower success for heavier payloads, while later launches demonstrated improved reliability across payload ranges.
- Interactive analytics demo in screenshots
  Interactive visualizations revealed site-specific and payload-dependent success patterns, enabling dynamic comparison of successful and failed launches. Geographic mapping confirmed that launch sites are strategically positioned near coastlines and distant from populated areas, supporting operational safety.
- Predictive analysis results
  Among the evaluated models, the Support Vector Machine (SVM) achieved the highest predictive performance, outperforming baseline models and demonstrating that mission outcomes can be predicted with reasonable accuracy using engineered features.

Section 2

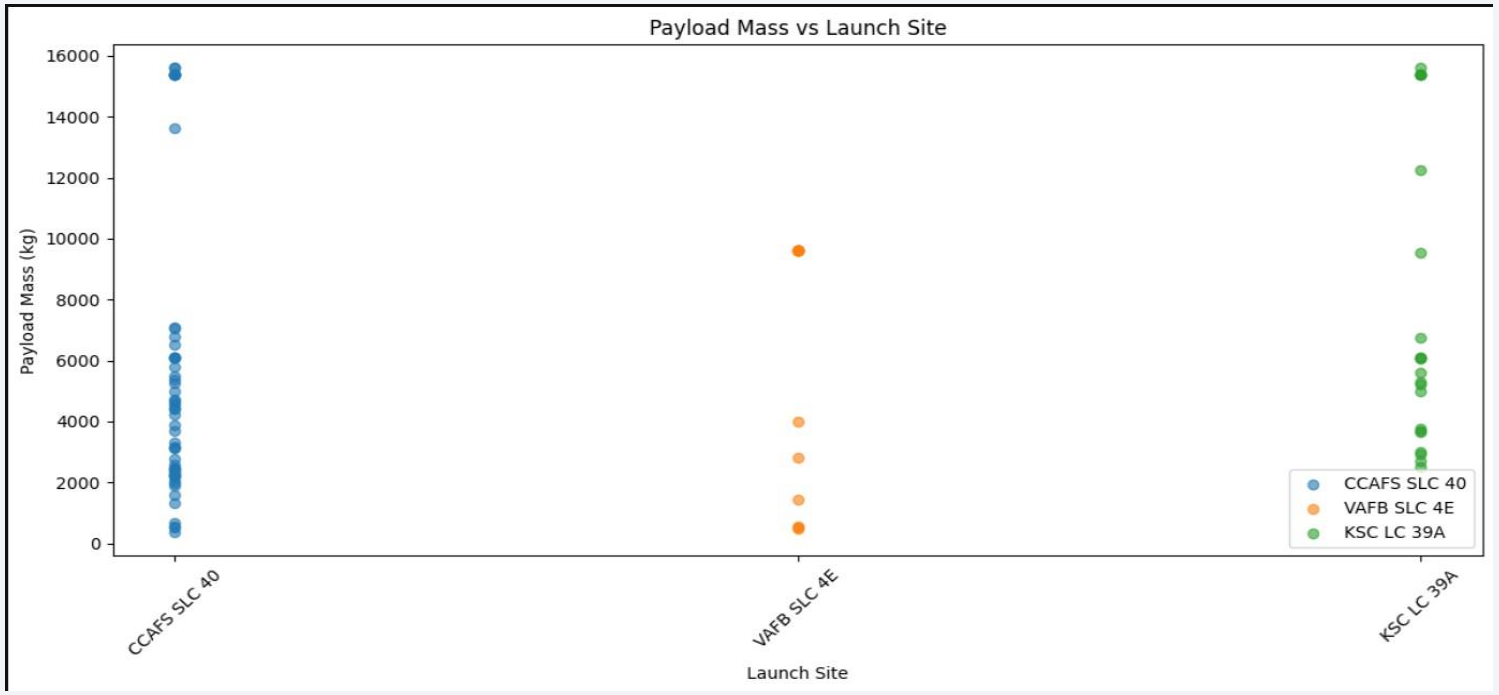
Insights drawn from EDA

Flight Number vs. Launch Site



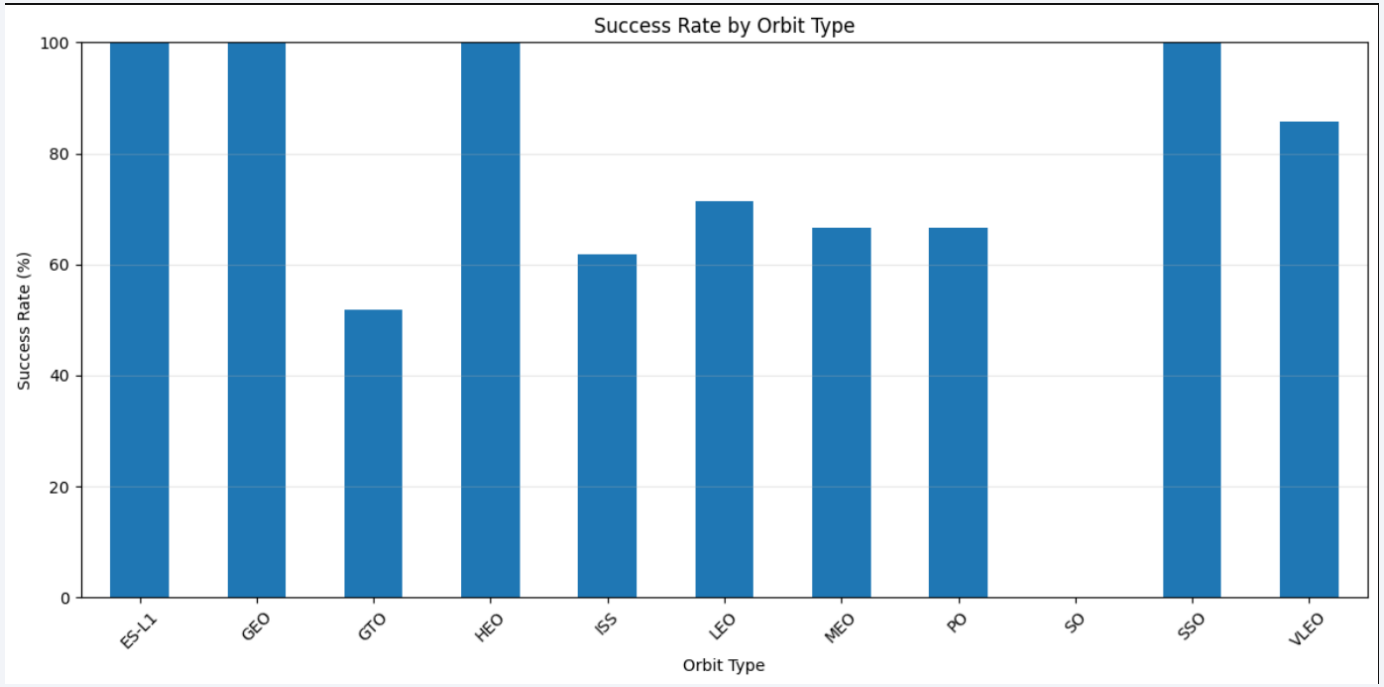
- This scatter plot correlates Payload Mass (kg) with Flight Number, distinguishing between successful (orange) and failed (blue) launches. The data reveals a learning curve where failures are concentrated in the early flight sequences, while later flights are predominantly successful, demonstrating improved reliability and operational maturity over time despite varying payload capacities.

Payload vs. Launch Site



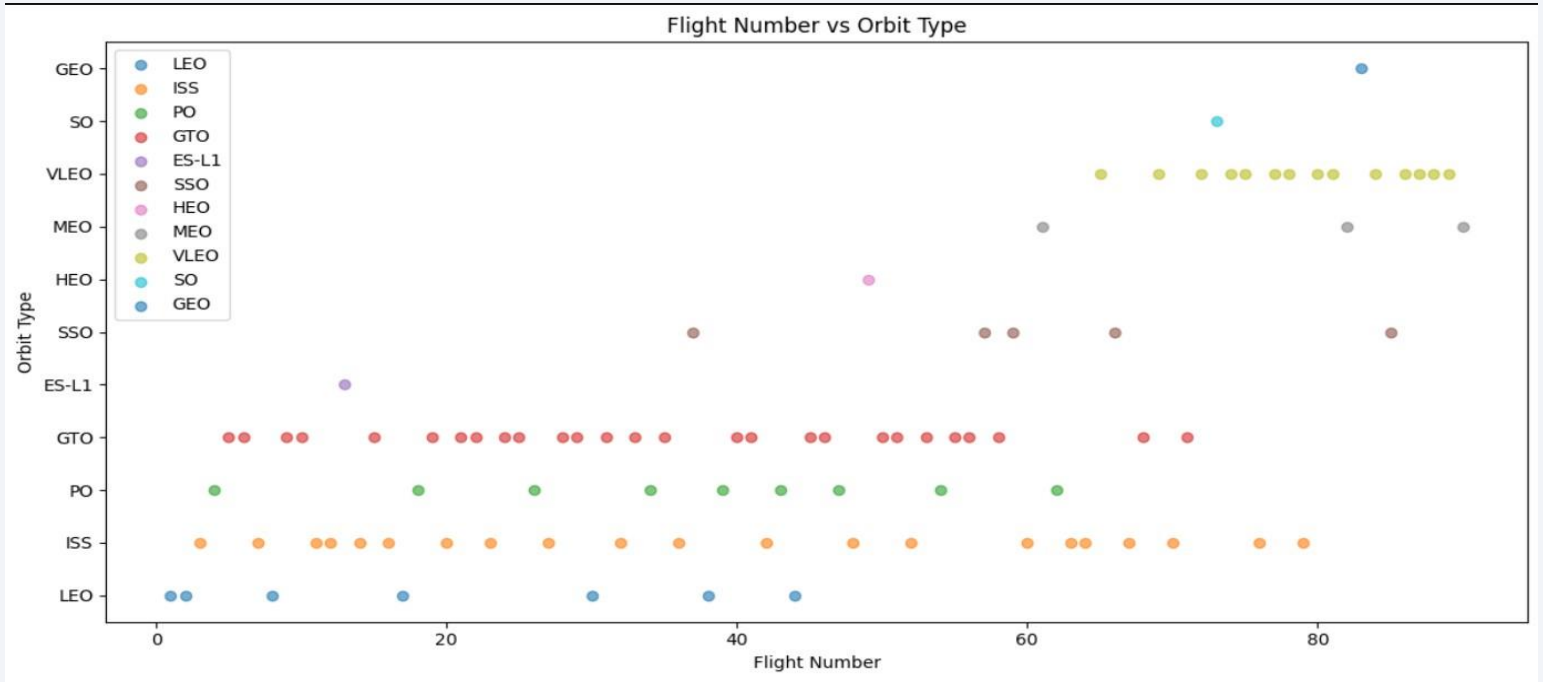
- The Payload Mass vs Launch Site chart compares payload weights across CCAFS SLC 40 (blue), VAFB SLC 4E (orange), and KSC LC 39A (green). The data reveals that payload mass is not uniformly distributed; CCAFS SLC 40 shows the highest frequency of launches across a wide range of masses, while KSC LC 39A tends to handle heavier payloads. This behavior indicates a correlation between launch site selection and specific payload weight requirements or mission profiles.

Success Rate vs. Orbit Type



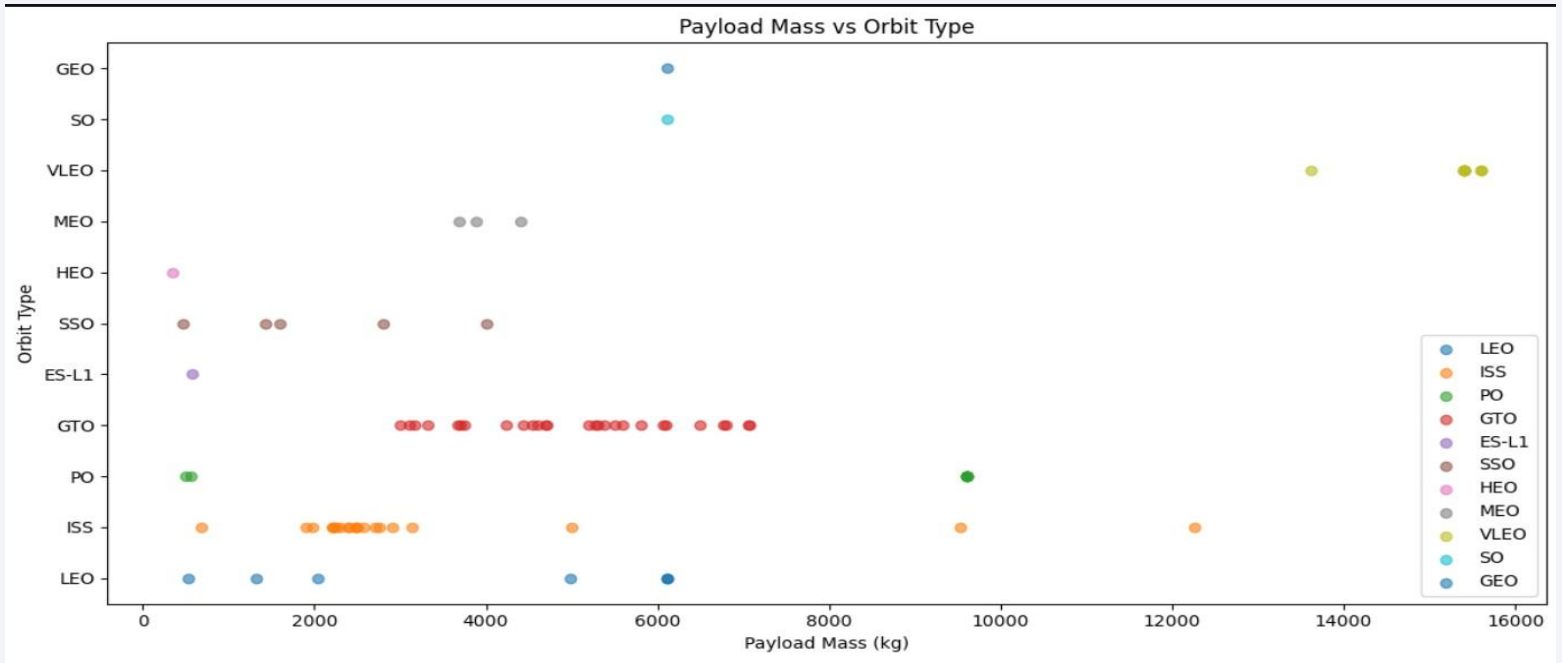
- The Success Rate by Orbit Type chart reveals that launch reliability varies drastically depending on the destination. While orbits like ES-L1, GEO, HEO, SSO, and VLEO boast near-perfect 100% success rates, others such as GTO, MEO, and Polar Orbit show significantly lower performance. This indicates that certain trajectories, particularly Geostationary Transfer Orbit (GTO), present distinct technical challenges that have historically resulted in higher failure probabilities compared to standard Low Earth or Highly Elliptical orbits.

Flight Number vs. Orbit Type



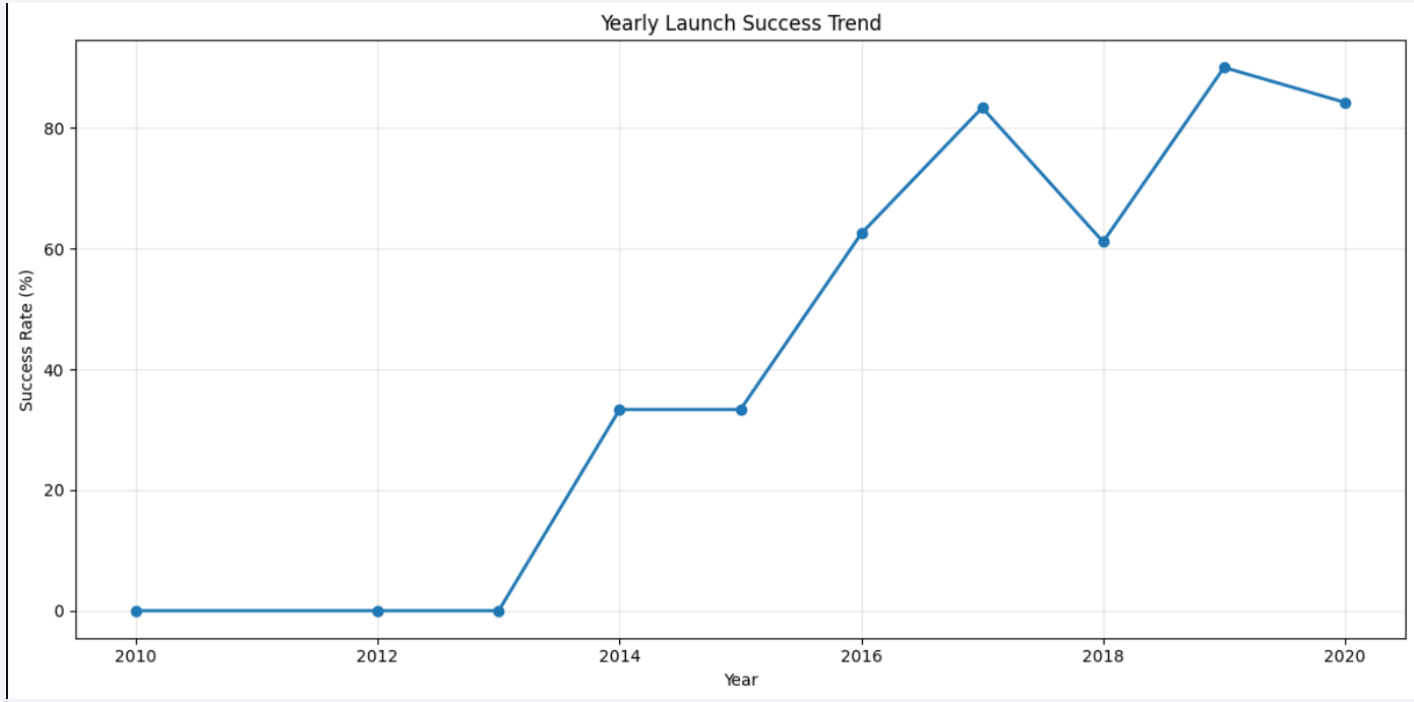
- The Flight Number vs Orbit Type chart illustrates the evolution of mission complexity over time. Early flights are primarily clustered around Low Earth Orbit (LEO) and ISS destinations, representing initial testing and verification. As the flight number increases, the data spreads vertically to include more demanding orbits like Geostationary Transfer Orbit (GTO), Sun-Synchronous (SSO), and Deep Space (ES-L1), indicating a progression from basic operations to supporting a wider variety of commercial and scientific orbital requirements.

Payload vs. Orbit Type



- The Payload Mass vs. Orbit Type chart illustrates the physical relationship between payload weight and destination. While Low Earth Orbit (LEO) accommodates a broad spectrum of masses from light to heavy, more energy-intensive orbits like Geostationary Transfer Orbit (GTO) show a tighter cluster of lighter payloads. This indicates that reaching distant orbits imposes stricter mass limitations compared to lower Earth orbits.

Launch Success Yearly Trend



- The Yearly Launch Success Trend chart illustrates a volatile yet upward trajectory in reliability from 2010 to 2020. Early years saw near-zero success rates, followed by a steady climb to 84% by 2017. Despite a dip in 2018, performance peaked at 91% in 2019 and stabilized at a high level in 2020, demonstrating a clear evolution from initial failure-prone launches to mature, high-success operations.

All Launch Site Names

- Unique Launch Sites

1 CCAFS LC-40

2 VAFB SLC-4E

3 KSC LC-39A

4 CCAFS SLC-40

- This dataset includes four distinct launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40. These sites represent two main geographical clusters: two variants at Cape Canaveral (CCAFS) and one each at Vandenberg (VAFB) and Kennedy Space Center (KSC). The separation of these sites reflects their specific operational roles, such as differentiating between east and west coast launches to accommodate various orbital inclinations and mission requirements.

Launch Site Names Begin with 'CCA'

Date Time (UTC)	Booster Version	Launch Site	Payload	Mass (kg)	Orbit	Customer	Landing Outcome
04-06-2010 18:45	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qual. Unit	0	LEO	SpaceX	Failure (parachute)
08-12-2010 15:43	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	LEO (ISS)	NASA (COTS) NRO	Failure (parachute)
22-05-2012 07:44	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	No attempt
08-10-2012 00:35	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	No attempt
01-03-2013 15:10	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	No attempt

- The provided dataset displays five specific records where the Launch_Site starts with "CCA" (Cape Canaveral Air Force Station). All five entries utilize the CCAFS LC-40 site and represent early SpaceX missions between 2010 and 2013. The data shows a progression from initial qualification tests with zero payload mass (B0003, B0004) to operational cargo missions to the ISS (B0005, B0006, B0007) carrying between 500kg and 677kg. While all missions were successful, landing outcomes varied from parachute failures to no landing attempts, reflecting the experimental nature of early recovery efforts.

Total Payload Mass

- The total payload mass carried by boosters launched for NASA (CRS) missions is 107,010 kg. This substantial aggregate weight reflects the significant volume of cargo and supplies transported to the International Space Station across multiple Commercial Resupply Services flights.

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is approximately 2,535 kg. This figure represents the mean weight of payloads delivered during the operational life of this specific Falcon 9 variant.

First Successful Ground Landing Date

- The first successful landing outcome on a ground pad was achieved on December 22, 2015. This milestone marked a historic achievement for SpaceX during the Falcon 9 Flight 20 mission, successfully landing the first stage at Cape Canaveral.

Successful Drone Ship Landing with Payload between 4000 and 6000

- The boosters that achieved a successful landing on a drone ship with a payload mass between 4000 and 6000 kg are F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2. These specific Falcon 9 Full Thrust variants successfully executed drone ship recoveries while carrying payloads within the specified mass range.

Total Number of Successful and Failure Mission Outcomes

- The mission data reveals a total of 100 Successful Missions and 1 Failed Mission. This overwhelming success rate underscores the high reliability and effectiveness of the launch operations during this period.

Boosters Carried Maximum Payload

- The booster versions that have carried the maximum payload mass are the F9 B5 series, specifically B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, and B1049.7. These 12 Falcon 9 Block 5 variants demonstrate the capability of the upgraded fleet to handle the heaviest payload requirements.

2015 Launch Records

- In 2015, there were two failed landing outcomes on drone ships, both occurring from the CCAFS LC-40 launch site. These incidents involved the F9 v1.1 B1012 booster in January (01) and the F9 v1.1 B1015 booster in April (04).

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **Landing Outcome Frequency (2010-06-04 to 2017-03-20)**

No attempt: 10

Success (drone ship): 5

Failure (drone ship): 5

Success (ground pad): 3

Controlled (ocean): 3

Uncontrolled (ocean): 2

Failure (parachute): 2

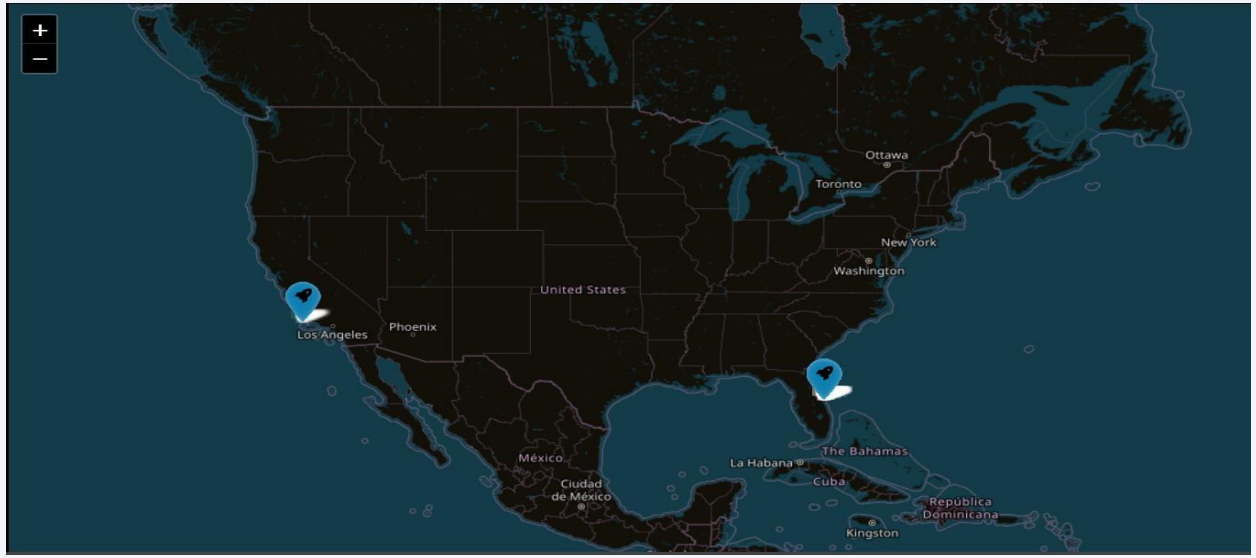
Precluded (drone ship): 1

Between June 4, 2010, and March 20, 2017, the most frequent landing outcome was No attempt (10 times), followed by an equal split of successes and failures on drone ships (5 each). There were 3 successful ground pad landings, while ocean landings resulted in 3 controlled and 2 uncontrolled outcomes, with only 2 parachute failures and 1 precluded drone ship landing.

Section 3

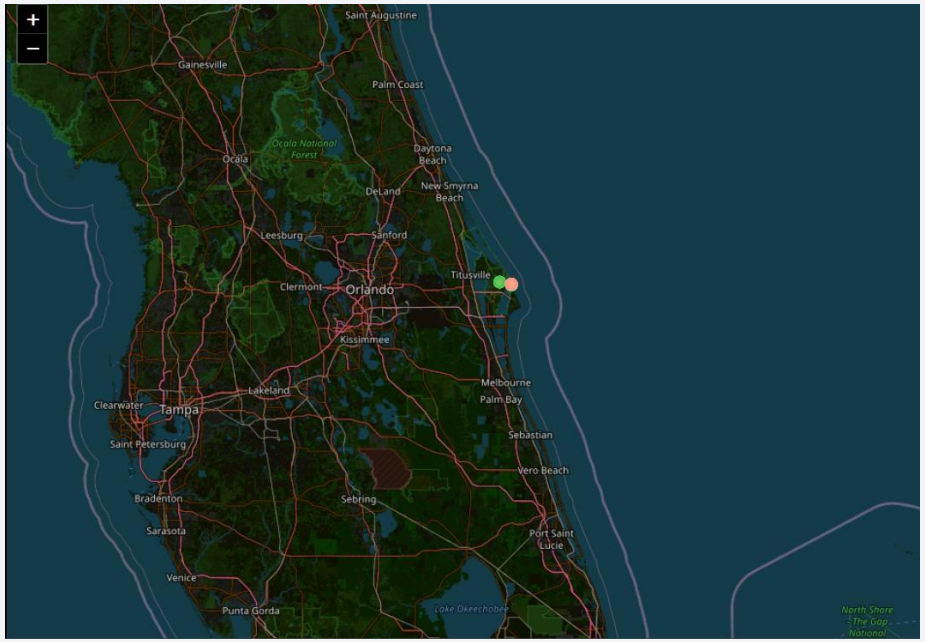
Launch Sites Proximities Analysis

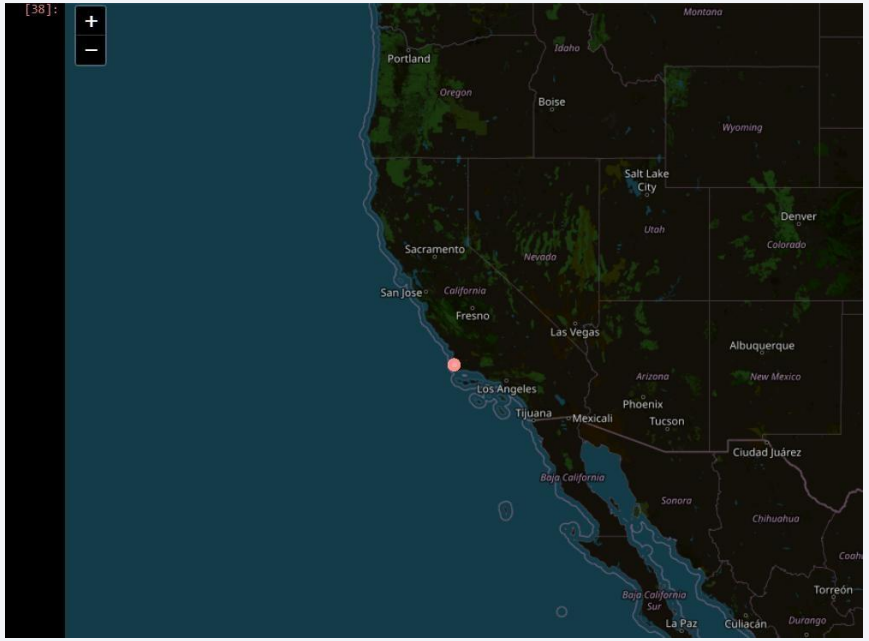
Global Distribution of SpaceX Launch Sites



- The map shows the global locations of SpaceX launch sites across the United States. Each marker represents a launch site and provides location information through interactive popups. This visualization highlights the geographic distribution of launch infrastructure and its proximity to coastlines.

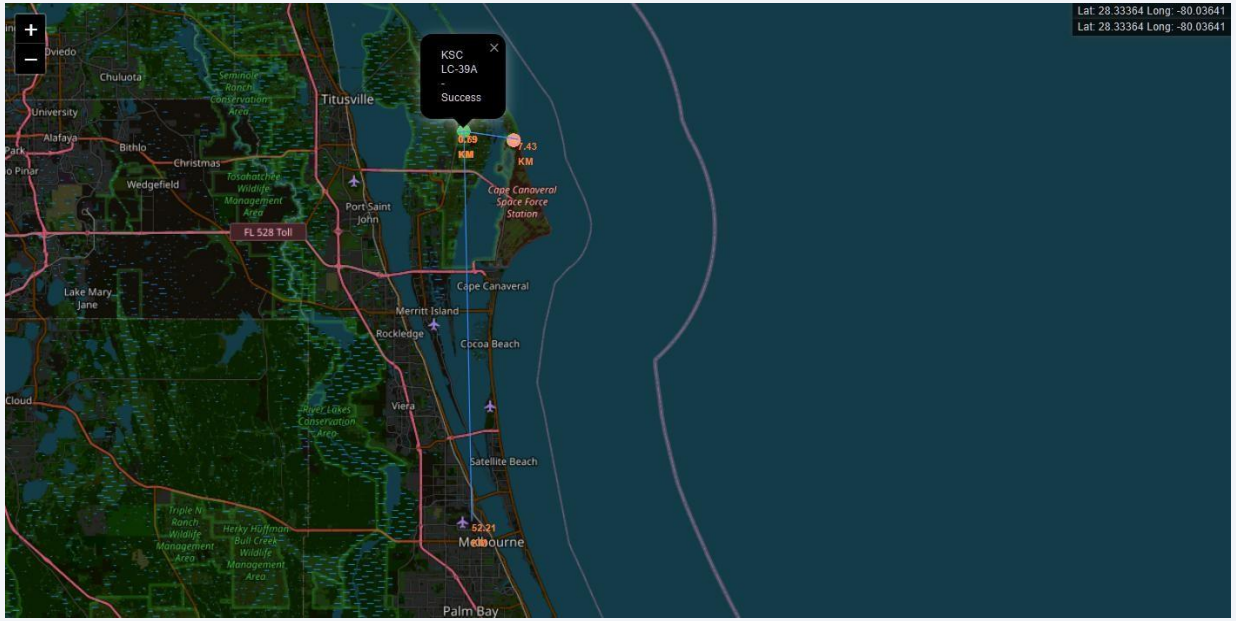
Launch Outcomes by Site

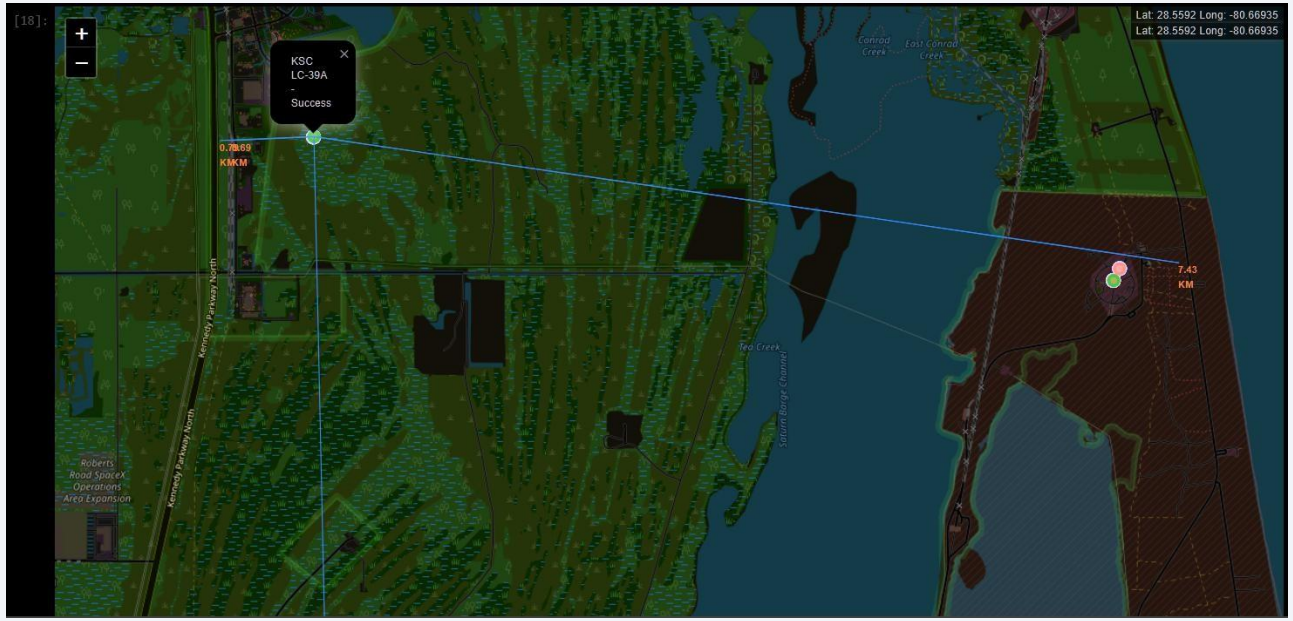




- This map visualizes launch outcomes using color-coded markers, where green indicates successful landings and red indicates failures. The visualization allows quick comparison of performance across launch sites and shows spatial patterns in mission success. Successes in Florida, while failures in California

Launch Site Proximity to Infrastructure and Coastline



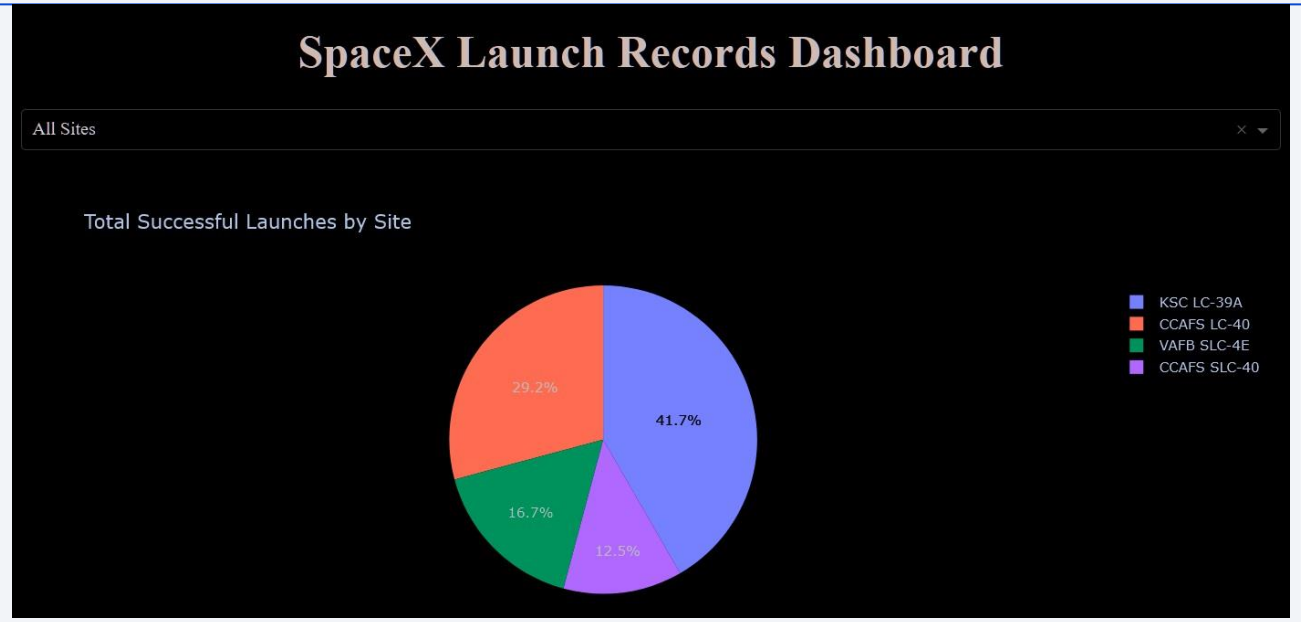


- These maps show the proximity of a successful launch site KSC LC-39A in Florida to key geographical features such as coastline (7.43 km), railways (690 m), and highways (800 m). Distance calculations are displayed to highlight safety, logistics, and accessibility considerations in launch site selection.

Section 4

Build a Dashboard with Plotly Dash

Launch Success Distribution by Site



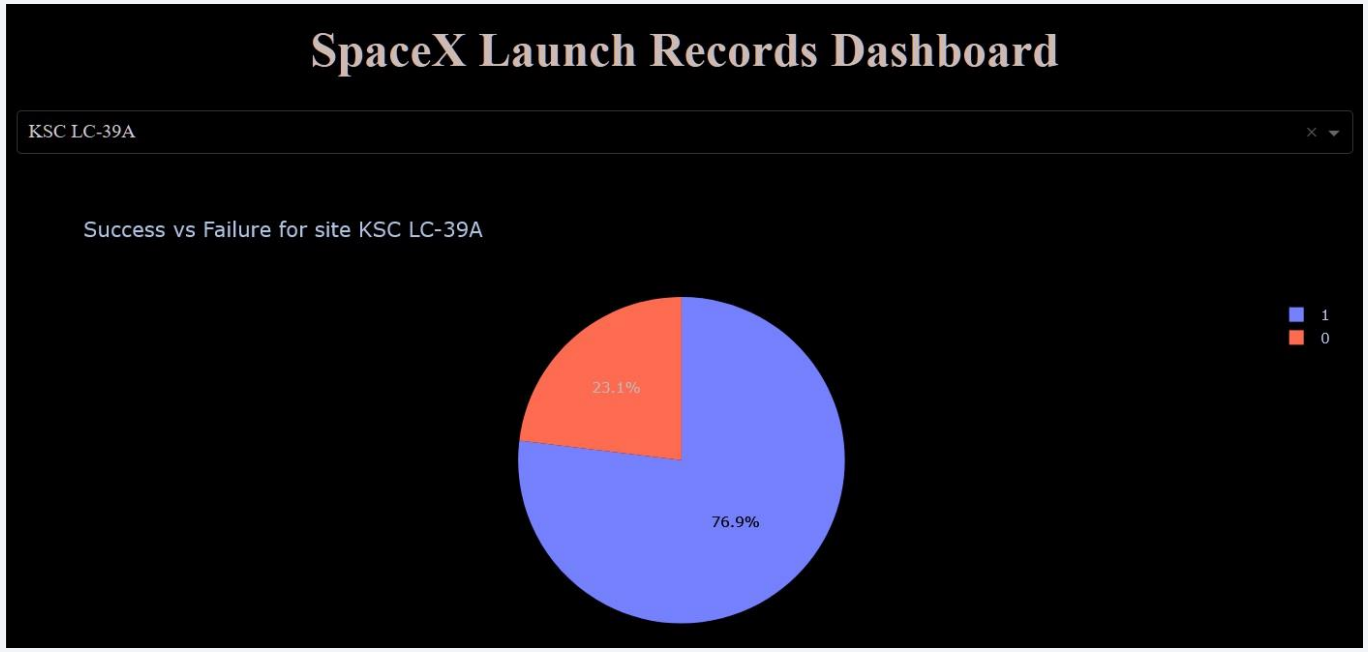
- Important elements

Dropdown menu (All Sites): Enables filtering by individual launch sites or viewing aggregated results.
Pie chart: Shows the proportion of successful launches contributed by each launch site.
Legend: Identifies each launch site and its corresponding share of successful launches.

- Key findings

  KSC LC-39A contributes the largest share of successful launches.
  CCAFS LC-40 and VAFB SLC-4E show moderate success contributions.
  CCAFS SLC-40 has the smallest proportion of successful launches among the sites.

Success vs Failure at KSC LC-39A



- Important elements

Launch site dropdown: Set to KSC LC-39A to isolate site-specific performance.

Pie chart: Visualizes the distribution of successful and unsuccessful launches.

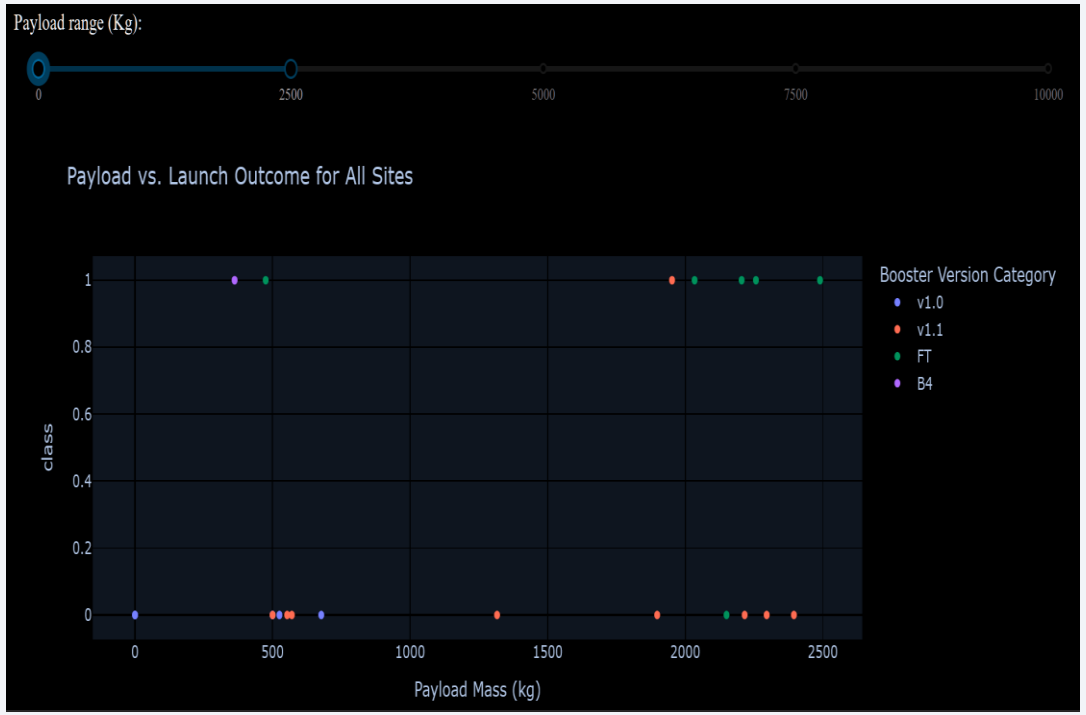
Color coding and legend: Clearly distinguish success and failure outcomes.

- Key findings

KSC LC-39A shows a high success rate (approximately 77%), indicating strong operational reliability.

The relatively smaller failure segment highlights improved launch consistency at this site compared to others.

Payload Mass vs Launch Outcome Across All Sites



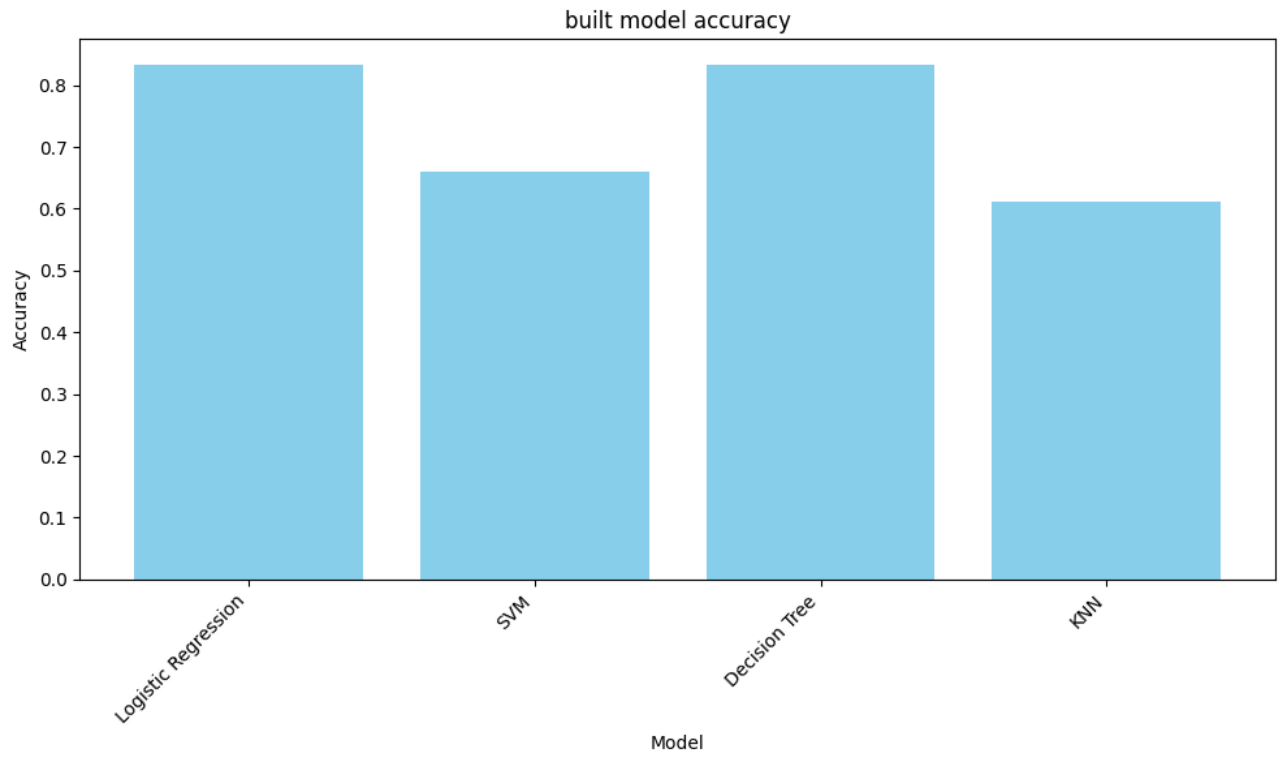
- The screenshots display scatter plots of Payload Mass (kg) versus Launch Outcome (class) for all launch sites.
- A payload range slider is used to filter the data, and multiple screenshots illustrate how the scatter plot updates for different payload ranges.
- Each data point represents a launch, colored by Booster Version Category.

- Important elements
  Payload range slider: Allows interactive filtering of launches by payload mass.
  Scatter plot:
  X-axis: Payload Mass (kg)
  Y-axis: Launch Outcome (0 = Failure, 1 = Success)
  Color-coded points: Represent different booster versions (v1.0, v1.1, FT, B4, B5).
  Multiple screenshots: Demonstrate how launch outcomes change as the payload range is adjusted.
- Key findings
  Successful launches (class = 1) are observed across a wide range of payload masses.
  Higher success concentrations appear in mid to high payload ranges, particularly with newer booster versions.
  Booster versions FT, B4, and B5 show a higher proportion of successful launches compared to earlier versions, indicating improved reliability over time.
  Failures (class = 0) are more frequent at lower payload ranges and among earlier booster versions.

Section 5

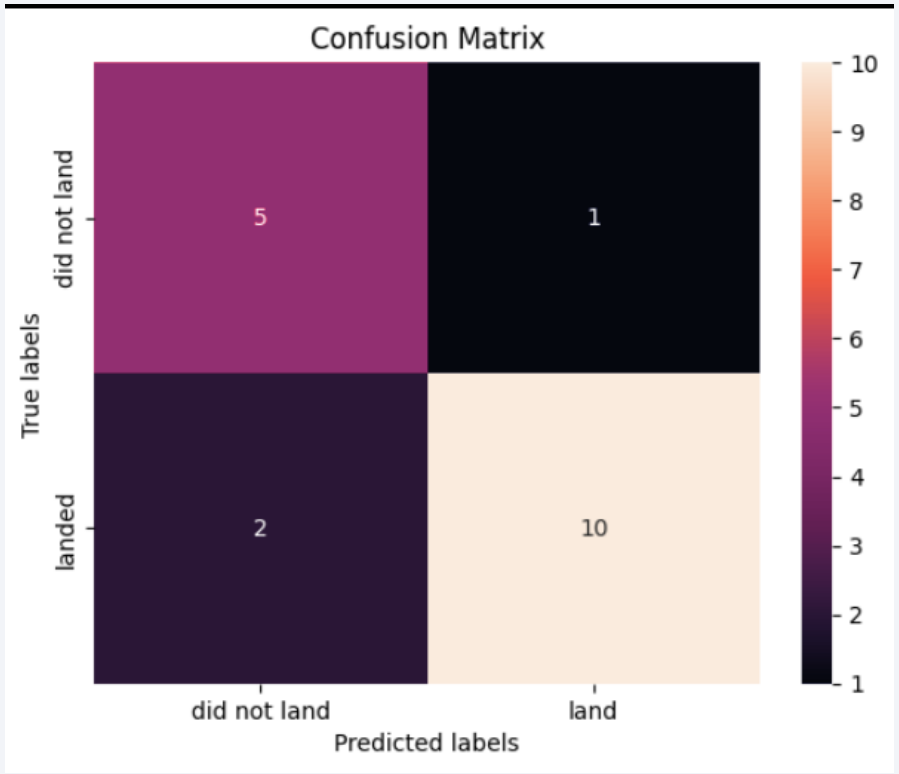
Predictive Analysis (Classification)

Classification Accuracy



- Decision Tree has the highest Accuracy (0.8333333334)

Confusion Matrix



- This confusion matrix displays the performance of the classification model. The True Labels (actual outcomes) are compared against the Predicted Labels.
- The model correctly predicted 10 "landed" cases (True Positives) and 5 "did not land" cases (True Negatives). However, it misclassified 1 "did not land" case as "landed" (False Positive) and 2 "landed" cases as "did not land" (False Negative). This indicates the model has a slight bias towards predicting "did not land," missing two actual successful landings.

Conclusions

- The analysis shows a clear improvement in SpaceX launch success rates over time, indicating effective technological evolution and learning from early mission failures.
- Launch success is influenced by a combination of factors, including payload mass, booster version, orbit type, and launch site location.
- Newer booster versions (FT, B4, and B5) consistently demonstrate higher landing success rates, reflecting improvements in design and reusability.
- Among all launch sites, KSC LC-39A exhibits the highest success ratio, suggesting optimized operational and infrastructural conditions.
- Payload mass alone does not determine mission outcome, as successful launches occur across a wide range of payload values.
- Interactive visualizations and dashboards enabled efficient exploration of complex launch data and supported evidence-based conclusions.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!